MIL-C-48458B(AR)
30 June 1981
SUPERSEDING
MIL-C-48458A(AR)
20 July 19

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732392 GCU ELECTRONICS

This specification is approved for use by the U.S. Army Armament Research and Development Comnand (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732395 GCU Electronics which is the A1 component of the Gunner's Control Unit, 117325000

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extend specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732392	Circuit Card Assembly: 11732392 GCU Electronics

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750428	Adapter, Gunner's Control Unit-A1
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter Plate Vib Horz-Vert Plane/ Shock Z
11821520	Automatic Test Assembly Control Processor Unit
11821865	Vibration/Shock Test Fixture - Gunner's Control Unit

Packaging Data Sheet

P11732392	Packaging of Circuit Card Assembly: 11732392

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should he obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The gunner's control unit (GCU) electronics circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732392 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following functions:

a. Ammo select switching logic and fail lamp diode logic for the GCU.
b. GCU common zero and zeroing signals switching logic.
c. Scaling resistors for various GCU outputs.

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a. Temperature	73°F + 18°F
b. Relative humidity	50 percent ± 30 percent
c. Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg,

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 Pin-to-pin resistance. The resistance between pins as listed in table II shall be as specified. (See 4.6.2.1)

3.3.2 Ammo select switching logic. With the interconnections (item 3) and power (item 1.1) of table I applied, establishment, in any item number order, of the table III items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power, signal sources and interconnections.

Item	Conditions	Characteristics	Connections
1.0	Power		Connected Between:
1.1	15.000 vdc	Tolerance: +0.150 vdc Ripple max 200 mv p-p	J1-27(+) and J2-38 (-) (rtn)
1.2	5.00 vdc	Tolerance: +0.05 vdc Ripple max 200 mv p-p	J1-5, -6, -7, -8, and J1-20, -48 (rtns)
1.3	-10.00 vdc	Tolerance: +0.05 vdc Ripple max 200 mv p-p	J1-5, -6, -7, -8, and J1-20, -48 (rtns)
2.0	Signal Sources		Connected Between:
2.1	5.0 vdc in series with a 100 kohm 1% 1/2 W resistor	Tolerance: ±0.2 vdc	J1-1, -2, -3, -4, -10, -11, -12, -13, and J1-20, -48 (rtns)
2.2	5.00 vdc in series with a 10 kohm 1% 1/2 W resistor	Tolerance: ±0.01 vdc	J1-56, -57, -58, -59, -60, -61, -62, -64, and J1-63 (rtns)
2.3	-5.00 vdc in series with a 10 kohm 1% 1/2 W resistor	Tolerance: ±0.01 vdc	J1-56, -57, -58, -59, -60, -61, -62, -64, and J1-63 (rtn)
3.0	Interconnections		Connected Between:
3.1			J1-37 with J1-81 and J1-89
3.2			J1-38 with J1-82

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:27Z
Check the source to verify that this is the current version before use.

TABLE II. Pin-to-pin resistance requirements.

Item	Between J1- Pins	Resistance
1	69 and 74	1.576 kilohms ± 0.8 ohms
2	69 and 73	1 kilohm ± 0.5 ohms
3	37 and 90	510 - 25 + 76 ohms
4	80 and 38	10 kilohms - 500 ohms + 1.5 Kilohms
5	83 and 38	10 kilohms - 500 ohms + 1.5 Kilohms
6	50 and 51	649 ± 6.5 ohms
7	51 and 52	221 ± 2.2 ohms
8	53 and 54	649 ± 6.5 ohms
9	52 and 54	221 ± 2.2 ohms
10	17 and 18	9.53 kilohms ± 95.3 ohms
11	21 and 22	73.2 kilohms ± 732 ohms
12	22 and 49	499 ± 4.99 kilohms
13	19 and 22	499 ± 4.99 kilohms
14	17 and 22	499 ± 4.99 kilohms
15	20 and 22	100 ± 1 kilohms
16	22 and 46	100 ± 1 kilohms
17	22 and 47	100 ± 1 kilohms
18	22 and 48	100 ± 1 kilohms
19	38 and 86	10 kilohms - 500 ohms + 1.5 kilohms
20	38 and 85	10 kilohms - 500 ohms + 1.5 kilohms

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

TABLE II. Pin-to-pin resistance requirements. - Continued

Item	Between J1- Pins	Resistance
21	38 and 77	20 - 1 + 3 kilohms
22	38 and 78	20 - 1 + 3 kilohms
23	32 and 37	510 - 25 + 76 ohms
24	37 and 42	510 - 25 + 76 ohms
25	27 and 30	20 - 1 + 3 kilohms
26	34 and 38	20 - 1 + 3 kilohms
27	36 and 38	20 - 1 + 3 kilohms
28	38 and 82	2.0 ohm max
29	37 and 89	2.0 ohm max
30	37 and 81	2.0 ohm max
31	21 and 30	70 kilohms min

TABLE III. Ammo select switching logic.

Item	Input Conditions at Pin J1-29 1/	Output Conditions at Pin J1-30 2/
1	1	0V
2	0	15V

1/ 1 = +15.000 ± 0.150 vdc (item 1.1, table I);
0 = 0.5 ± 0.5 vdc (Return of item 1.1, table I).

2/ 15v = 15.0 ± 0.5 vdc; 0v = 0.0 ± 0.5 vdc.

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:27Z
Check the source to verify that this is the current version before use.

3.3.3 GCU common zero and zeroing signals switching logic. With the interconnections (item 3) and the 5 vdc signal (item 2.1 of table I applied, and 1.5 vrms signal (item 3.3) of table I applied, establishment, in any item number order, of the table IV items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.3)

3.3.4 Fail lamp diodes. With the interconnections (item 3 of table I) applied and with the 5 vdc signal (item 2.2 of table I) applied in any order to pins J1-56 through J1-62 and to J1-64, the output at J1-63 shall be 060 ± 0.20 vdc. With the interconnections (item 3 of table I) applied and with the -5 vdc signal (item 2.3 of table I), applied in any order to pins J1-56 through J1-62 and to J1-64, the output at J1-63 shall be - 5.00 ±0.01 vdc. (See 4.6.2.4)

TABLE IV. GCU common zero and zeroing signals switching logic.

Item	Input Conditions at Pins 1/, 3/				Output Conditions at Pins 2/, 3/			
	J1-5	J1-6	J1-7	J1-8	J1-4 and J1-12	J1-3 and J1-11	J1-2 and J1-10	J1-1 and J1-13
1	1	0	0	0	0v	5v	5v	5v
2	0	1	0	0	5v	0v	5v	5v
3	0	0	1	0	5v	5v	0v	5v
4	0	0	0	1	5v	5v	5v	0v

1/ 1 = +5 vdc (table I, item 1.2).
0 = -10 vdc (table I, item 1.3).

2/ 5v = table I, item 2.1, Measured Voltage ±0.05 vdc.
0v = 0.00 ± 0.05 vdc.

3/ Returns are J1-20 and -48.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 50 ± 5 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 125.0 ± 12.5 g for a duration of 0.6 ± 0.2 msec applied in each direction along three mutually perpendicular axes. The tolerance shall be ± 0.2 msec on the first shock for each direction set up and ± 0.1 msec for subsequent shocks for each direction set up when the specified pulse duration is 1.0 msec or less. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 1. Duration of exposure shall be not less than 80 minute in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specitied herein after exposure to storage temperatures ranging from -65°F. to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

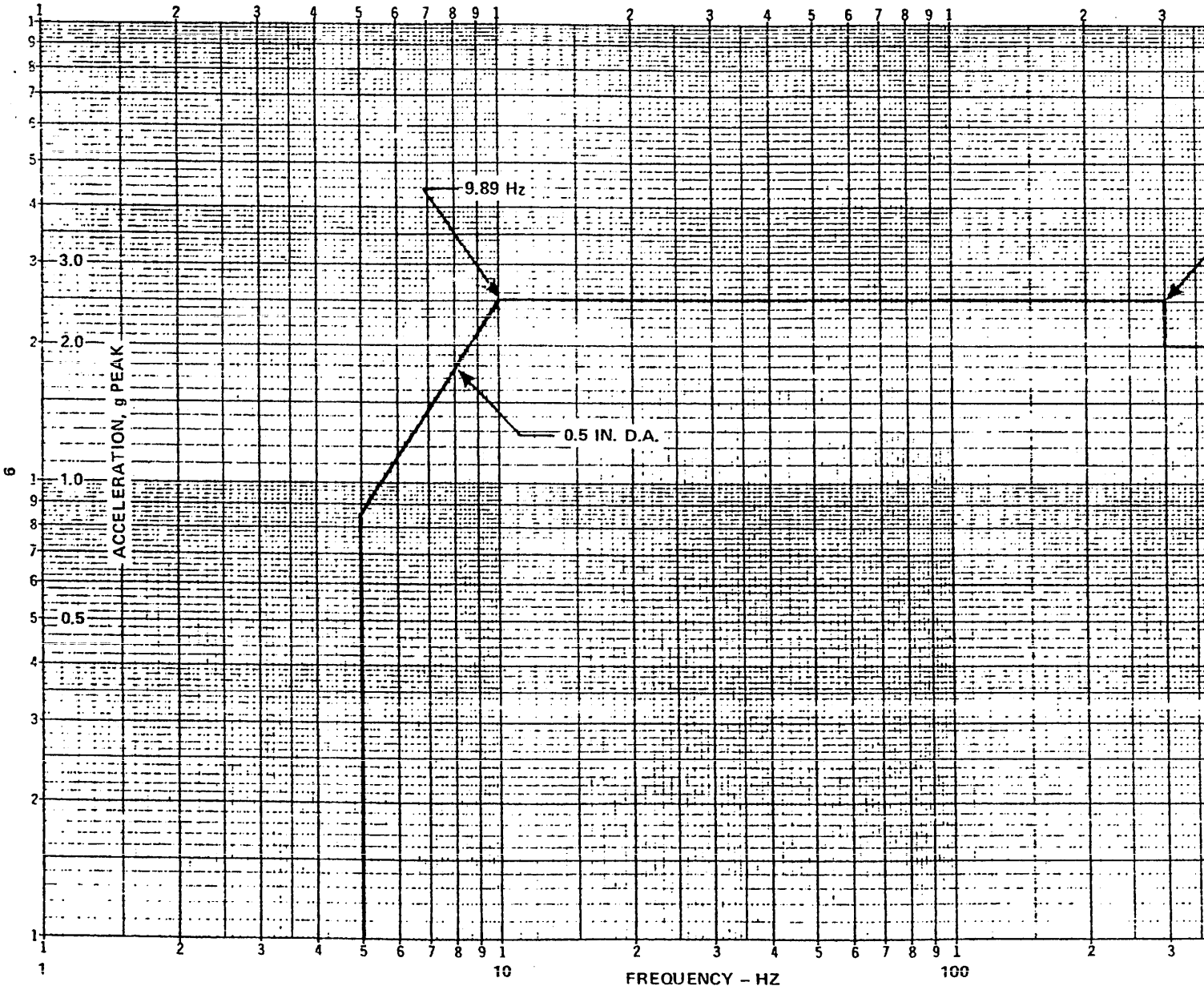


FIGURE 1. Vibration profile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables V, VI and VII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon an failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

TABLE V. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	Pin-to-pin resistance	3.3.1	4.6.2.1
102	Ammo select switching logic	3.3.2	4.6.2.2
103	GCU common zero and zeroing signals switching logic	3.3.3	4.6.2.3
104	Fail lamp diodes	3.3.4	4.6.2.4

TABLE VI. General tests.

Item	Characteristic	Requirement	Test Procedures
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE VII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly.	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VI.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VII shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet the specified requirements, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of M1L-STD-105. All the tests in table VI and only the test of item 307 in table VII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected, to all the tests in table V and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732392.

TABLE VIII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750428	Adapter, Gunner's Control Unit-Al
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter Plate Vib Horz-Vert
5	11821520	Automatic Test Assembly Control Processor Unit
6	11821865	Vibration/Shock Test Fixture-Gunner's Control Unit

4.6.2 Performance test.

4.6.2.1 Pin-to-pin resistance. Use items 1, 2 and 5 of able VIII or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 Amno select switching logic. Use items 1, 2 and 5 of table VIII or suitable equipment to verify the requirements of 3.3.2.

4.6.2.3 GCU common zero and zeroing signals switching logic. Use items 1, 2 and 5 of table VIII or suitable equipment to verify the requirements of 3.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

4.6.2.4 Fail lamp diodes. Use items 1, 2 and 5 of table VIII or suitable equipment to verify the requirements of 3.3.4.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 6 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure 1, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 50 ±5 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.1.2 High intensity. Use items 3, 4 and 6 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.1, procedure IV, figure 516.2-2. Apply three halt sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 125 ± 12.5 g with a time duration of 0.6 ± 0.2 msec measured at the 10 percent amplitude points. The tolerance shall be ± 0.2 msec on the first shock for each direction set up and ±0.1 msec for subsequent shocks for each direction set up when the specified pulse duration is 1.0 msec or less. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for qualify conformance inspection.

4.6.3.2.1 Vibration A. Use items 3, 4 and 6 of table VIII or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table III.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

4.6.3.3 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table V shall be applied.

4.6.3.4 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table V shall be applied.

4.6.3.5 Temperate cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 3. At the conclusion of this test, subject the assembly to the performance tests of table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

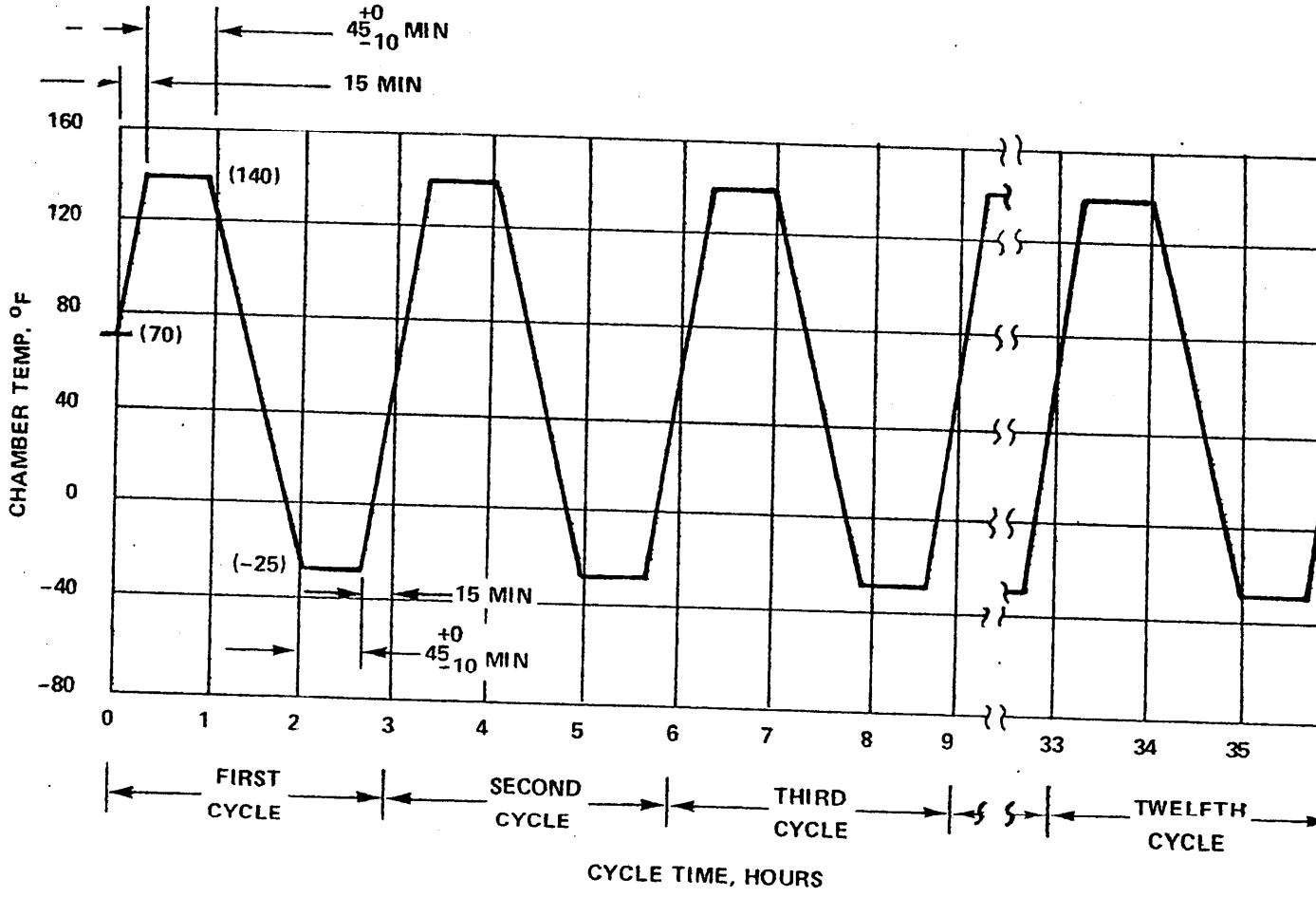


FIGURE 2. Temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732392. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The GCU Electronics - A1 circuit card assembly is used in the Gunner's control Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examinations as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging, and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48458B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for Publication.

Custodian:
Army-AR

Preparing Activity:
Army - AR
Project Number: 1220-A28

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use i submitting comments and suggestions for improvements. All users of military standardization documents are invited [illegible] suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STA[illegible]*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, w too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:27Z
Check the source to verify that this is the current version before use.